
Beyond Fitness Curves: Visualizing Population Dynamics in Neuroevolutionary Search

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Abstract

1 Neuroevolutionary algorithms maintain entire populations of candidate solutions
2 across generations, yet their dynamics are typically analyzed only through scalar
3 fitness metrics, which discard emergent geometric structure in the solution space.
4 We propose an analysis approach based on temporally aligned UMAP embeddings
5 and population velocity fields, with a post-hoc alignment procedure that reduces
6 arbitrary drift between successive projections, making temporal comparisons con-
7 sistent. Across two benchmarks with qualitatively distinct search regimes—Make
8 Moons ($d_w = 48$) and CIFAR-10 ($d_w \approx 197,000$)—we observe coherent popula-
9 tion flows associated with convergence basins, spatial localization of high-fitness
10 individuals, and task-dependent multimodal structure invisible in scalar metrics.
11 We show that these structures are largely obscured in unaligned projections, while
12 embedding-space velocity fields expose interpretable dynamic regimes through
13 spatial divergence patterns, suggesting that population geometry provides a com-
14plementary lens for understanding exploration, convergence, and multimodality in
15 neuroevolution.

1 Introduction

17 Neuroevolutionary algorithms [25, 26] optimize neural network parameters through population-
18 based search, maintaining a diverse set of candidate solutions across generations. Despite their
19 widespread use, our understanding of their internal dynamics remains largely limited to scalar
20 summaries—mean fitness curves and best-individual trajectories—that collapse the population’s full
21 geometry into a single number per generation.

22 This is a significant blind spot. A population of neural networks is a cloud of points in a high-
23 dimensional weight space, and its evolution—convergence basins, exploratory spread, multimodal
24 structure—constitutes rich information about how search is proceeding. Scalar fitness curves may
25 not distinguish between a population converging toward a single attractor and one that is simultane-
26ously exploring multiple disconnected fitness peaks.

27 **Our approach.** We propose analyzing neuroevolution through temporally aligned low-
28 dimensional embeddings of evolving weight populations, projecting each generation into 2D via
29 UMAP [17, 16] with a post-hoc alignment step, then deriving *population velocity fields* that summa-
30rize average individual displacement across generations (Section 3).

31 **Contributions.** Using this methodology, we identify three population-level geometric phenomena
32 that have no analogue in single-network trajectory analysis, where there is no selection pressure, no
33 competition between individuals, and no emergent multimodality across a population:

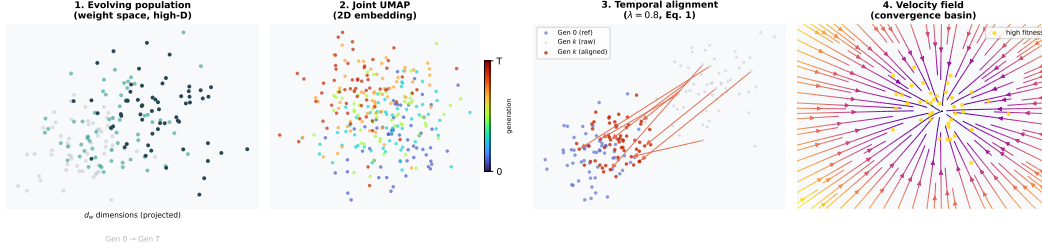


Figure 1: **Analysis pipeline.** At each generation, the neuroevolutionary population (a cloud of weight vectors) is projected jointly into 2D via UMAP. A post-hoc alignment step (Eq. 1, $\lambda = 0.8$) reduces arbitrary between-generation drift, making successive projections directly comparable. From the aligned per-generation embeddings we derive a *population velocity field*: for each cell of a 22×22 grid, the mean individual displacement from generation k to $k+1$ is computed and rendered as a streamline. The resulting visualization reveals convergence basins, multimodal structure, and fitness localization that are invisible in scalar fitness curves. This pipeline adapts the projection-trajectory vector field methodology of [1] from single-network hidden activations to evolving weight populations.

- 34 • **(i) Coherent flow structure.** Population trajectories exhibit organized directional flow
35 rather than isotropic stochastic spread, with identifiable convergence basins visible across
36 multiple independent random seeds. This collective structure is driven by selection and
37 cannot be observed by tracking a single network.
- 38 • **(ii) Selection-driven fitness localization.** High-fitness individuals concentrate in compact,
39 spatially coherent regions of the embedding. This spatial clustering emerges from selection
40 pressure acting across the population—a phenomenon absent when visualizing a single
41 network’s internal states.
- 42 • **(iii) Distinct flow regimes across benchmarks.** The two benchmarks exhibit qualitatively
43 different population geometries: Make Moons ($d_w = 48$, hidden=16) shows compact
44 single-basin flow, whereas CIFAR-10 ($d_w \approx 197,000$, hidden=64) shows persistent multi-
45 attractor structure. We report this as an empirical observation; we do not attribute it to any
46 single factor (task complexity, dimensionality, mutation scale, or architecture) given that
47 all differ between benchmarks (Section 6). This geometric difference is invisible in scalar
48 fitness curves.

49 We additionally validate that these structures are not artifacts of the UMAP projection by (a) compar-
50 ing aligned vs. unaligned projections and alternative dimensionality reduction baselines, (b) ablating
51 the alignment strength, and (c) verifying that known synthetic cluster structure is faithfully recovered
52 in UMAP embeddings.

53 2 Background

54 2.1 Neuroevolution

55 Neuroevolutionary algorithms represent neural network weights as genomes and apply evolutionary
56 operators—mutation, selection, occasionally crossover—to optimize a fitness function evaluated on
57 the network’s behavior [5, 25]. A central design choice is maintaining a *population* of individuals,
58 enabling implicit parallelism and resistance to local optima. We focus on mutation-only evolution
59 with elite selection, a simple but effective baseline used in several recent large-scale studies [26, 24].

60 2.2 Dimensionality Reduction for High-Dimensional Data

61 UMAP [17] is a nonlinear dimensionality reduction method that preserves local and global structure
62 through a graph-based approximation to manifold learning. Unlike PCA, it can represent non-linear
63 manifold structure; unlike t-SNE [27], it scales better to large datasets and preserves more global

64 geometry. Both have been used to visualize neural network weight spaces [15, 7] and hidden ac-
 65 tivations [21, 1, 8], but not for tracking *populations* of independent networks across evolutionary
 66 generations.

67 2.3 Temporal Alignment of Projections

68 A fundamental challenge in applying dimensionality reduction to sequential data is that each in-
 69 dependent projection may differ by an arbitrary rotation, reflection, or translation, making direct
 70 comparison across time steps unreliable [16]. We address this with a *reference-anchored interpola-*
 71 *tion*: all generations are projected jointly into a single UMAP embedding (ensuring a shared global
 72 coordinate system), then each generation’s coordinates are interpolated toward the first generation’s
 73 embedding:

$$\tilde{\mathbf{Z}}_k = (1 - \lambda) \mathbf{Z}_k + \lambda \mathbf{Z}_{\text{ref}}, \quad (1)$$

74 where $\mathbf{Z}_k \in \mathbb{R}^{n \times 2}$ is the embedding of generation k , $\mathbf{Z}_{\text{ref}}$ is the reference (gen 0), and $\lambda \in [0, 1]$
 75 controls alignment strength ($\lambda = 0$: raw embedding; $\lambda = 1$: collapse to reference). We set $\lambda = 0.8$
 76 based on an ablation study (Section 4.3).

77 We use joint embedding with post-hoc interpolation rather than Aligned UMAP [16] because it pre-
 78 serves a single global coordinate system without requiring consecutive-generation correspondence
 79 during graph construction.

80 3 Methodology

81 3.1 Neuroevolution Setup

82 We evolve one-hidden-layer MLPs using mutation-only evolution with elite selection. At each gen-
 83 eration, the top $\alpha = 20\%$ of individuals (by fitness) are retained; the remainder are replaced by
 84 mutated copies of the elite, where mutations are sampled $\delta w \sim \mathcal{N}(0, \sigma^2)$ independently per weight.
 85 We study two benchmarks (Table 1):

Table 1: Benchmark configurations. d_w = total weight dimensions; σ = mutation standard deviation.

Benchmark	n pop	Gens	Hidden	σ	d_w	PCA pre-reduction
Make Moons	50	80	16	0.11	48	None
CIFAR-10	20	40	64	0.05	$\approx 197\text{k}$	50 dims

86 **Make Moons** is a 2D binary classification task from `sklearn` [19] with noise $\sigma = 0.1$. Fitness
 87 is negative cross-entropy loss on the full dataset. **CIFAR-10** [11] is a 10-class image classification
 88 task. Due to the high weight dimensionality ($d_w \approx 197,000$), we apply PCA to 50 dimensions
 89 before computing UMAP embeddings, following standard practice for high-dimensional weight
 90 visualization [15].

91 We deliberately study small-scale settings (one-hidden-layer MLPs, populations of 20–50) to enable
 92 rigorous multi-seed analysis and synthetic validation at tractable compute cost; scaling to larger
 93 populations and deeper architectures is left to future work. Code and instructions to reproduce all
 94 experiments are available at <https://anonymous.4open.science/r/neurips-2026-9729>.

95 3.2 Temporally Aligned UMAP Embeddings

96 All individuals across all generations are concatenated into a single matrix and projected jointly by
 97 UMAP (`neighbors = 15`, `min_dist = 0.1`), ensuring a consistent global coordinate system. Generation
 98 labels are preserved for color-coding. The alignment of Equation 1 is then applied point-wise using
 99 the first generation as reference.

100 3.3 Population Velocity Fields

101 From the aligned per-generation embeddings, we construct a 2D velocity field summarizing popu-
 102 lation flow. The embedding space is partitioned into a 22×22 grid; for each cell, we compute the

mean displacement vector of individuals that are present in both generation k and $k + 1$ and whose centroid falls in that cell. The resulting field is smoothed with a 3×3 box kernel before visualization.

3.4 Metrics

Embedding spread (σ_1, σ_2) is the standard deviation of aligned embedding coordinates along each axis, measuring how dispersed the population is in projection space. Larger spread indicates greater dispersion; whether this reflects multimodality or slow convergence is disambiguated by the velocity field visualization (Section 3.3).

Temporal coherence measures how consistently the arrangement of individuals is preserved across successive projections:

$$TC = \frac{1}{T-1} \sum_{k=1}^{T-1} \left| \text{Pearson}(\tilde{\mathbf{Z}}_k, \tilde{\mathbf{Z}}_{k+1}) \right|, \quad (2)$$

where Pearson correlation is taken over flattened embedding vectors. Absolute value is used because UMAP may produce arbitrary reflections between independently projected generations.

Convergence rate is the first generation at which mean fitness reaches 95% of its final improvement: $g^* = \min\{g : \bar{f}_g \geq \bar{f}_0 + 0.95(\bar{f}_T - \bar{f}_0)\}$.

4 Results

4.1 Exp 1: Multi-Seed Robustness

To verify that observed structures are not artifacts of a particular random initialization, we run both benchmarks with 5 independent seeds (42, 123, 7, 31, 99) and compare the resulting embeddings and metrics (Table 2).

Table 2: Multi-seed robustness (mean $\pm$ std, 5 seeds). Spread σ_1 is the standard deviation of the aligned UMAP embedding along the first axis; convergence rate is the generation at which 95% of fitness improvement is achieved. Welch’s t -tests (Make Moons vs. CIFAR-10): σ_1 : $t(5.0) = -5.84$, $p = 0.0021$, $d = 3.69$; convergence: $t(6.7) = 5.37$, $p = 0.0012$, $d = 3.40$.

Benchmark	Embedding spread σ_1	Final fitness	Convergence (gen)
Make Moons	1.97 ± 0.08	-0.171 ± 0.006	47 ± 6
CIFAR-10	2.62 ± 0.22	-2.490 ± 0.022	20 ± 10

Make Moons shows compact, consistent embedding spread ($\sigma_1 = 1.97 \pm 0.08$) across all 5 seeds, with very low variance confirming that the near-unimodal structure is a robust property of the search process. CIFAR-10 exhibits substantially larger spread ($\sigma_1 = 2.62 \pm 0.22$), consistent with a more dispersed, multimodal population geometry. The difference is statistically significant despite the small sample: Welch’s t -test yields $t(5.0) = -5.84$, $p = 0.0021$, Cohen’s $d = 3.69$ (large effect). Convergence is faster for CIFAR-10 (20 ± 10 generations vs. 47 ± 6 ; $t(6.7) = 5.37$, $p = 0.0012$, $d = 3.40$) but to a substantially lower fitness level (-2.49 vs. -0.17), suggesting rapid collapse into one of several local basins rather than sustained global exploration. The high convergence variance for CIFAR-10 (± 10 gens) further reflects seed-dependent dynamics in the multimodal landscape. The velocity fields (Figure 2) confirm these differences visually: Make Moons shows radial convergence toward a single compact region, while CIFAR-10 shows branching flows toward multiple coexisting sink regions across all seeds.

4.2 Exp 2: Projection Baselines

We compare four projection methods on the Make Moons benchmark: PCA, t-SNE [27], UMAP unaligned, and UMAP aligned ($\lambda = 0.8$). For each method, we visualize generation coloring, fitness coloring, and the derived velocity field.

PCA recovers a two-moons-like structure in weight space (Figure 3, top row), reflecting input-space geometry that bleeds into early-layer weights; this is static dataset-induced structure, not search

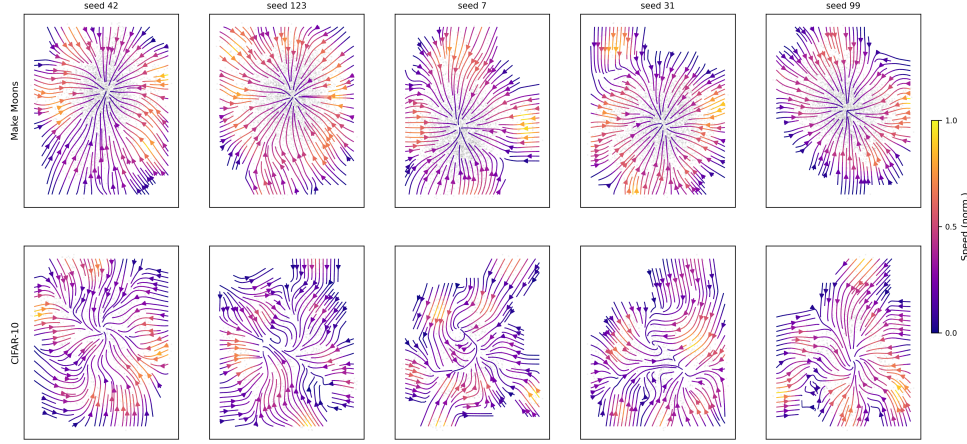


Figure 2: **Multi-seed reproducibility of population flow patterns.** 2 rows (Make Moons top, CIFAR-10 bottom) $\times$ 5 columns (seeds 42, 123, 7, 31, 99). Make Moons consistently shows radial convergence toward a single attractor ($\sigma_1 = 1.97 \pm 0.08$; $p = 0.0021$ vs. CIFAR-10, Welch’s t -test). CIFAR-10 reproduces 4–5 coexisting attractors per seed ($\sigma_1 = 2.62 \pm 0.22$); attractor locations vary across seeds, but the multi-sink topology is consistent. CIFAR-10 seed 31 (column 4) has an anomalously low scalar convergence rate (gen 3); see Table 5[†] for explanation. Together these support claims (i) and (iii).

dynamics. PCA conflates generations without revealing temporal flow. t-SNE fragments the embedding into many small disconnected clusters (attractor count = 298 by DBSCAN), making velocity fields uninterpretable. UMAP unaligned shows more structure (attractor count = 64) but exhibits arbitrary between-generation drift that distorts velocity fields. UMAP aligned ($\lambda = 0.8$) produces the most compact and temporally coherent embedding (attractor count = 21, spread $\sigma_1 = 1.88$), with smooth velocity fields that clearly reveal convergence dynamics. (The spread here uses seed 42 only; the cross-seed mean is 1.97 ± 0.08 , Table 2.)

4.3 Exp 3: Alignment Strength Ablation

We sweep $\lambda \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$ on the Make Moons benchmark and measure temporal coherence and structural compactness. Results are shown in Figure 4 and Table 3.

Table 3: Exp 3: alignment ablation on Make Moons. Temporal coherence and embedding spread as a function of λ .

λ	Temporal coherence	Attractor count	Spread σ_1
0.0	−0.010	10	8.58
0.2	0.005	10	6.88
0.4	0.045	10	5.20
0.6	0.169	7	3.53
0.8	0.587	5	1.88
1.0	1.000	10	0.93

Temporal coherence increases monotonically with λ , reaching TC = 1.0 at $\lambda = 1.0$ (all embeddings collapse to the reference by construction). However, $\lambda = 1.0$ eliminates all temporal variation and compresses the embedding ($\sigma_1 = 0.93$), destroying interpretable structure. The value $\lambda = 0.8$ provides the best trade-off: high coherence (TC = 0.587) with meaningful spread ($\sigma_1 = 1.88$), and velocity fields that visually reveal distinct convergence dynamics (Figure 4); the full TC vs. λ curve is shown in Appendix A.5 (Figure 6). Attractor count (Table 3) is provided as an ordinal indicator only; as noted in Appendix A.3, it is sensitive to embedding density and should not be interpreted as a precise geometric measurement.

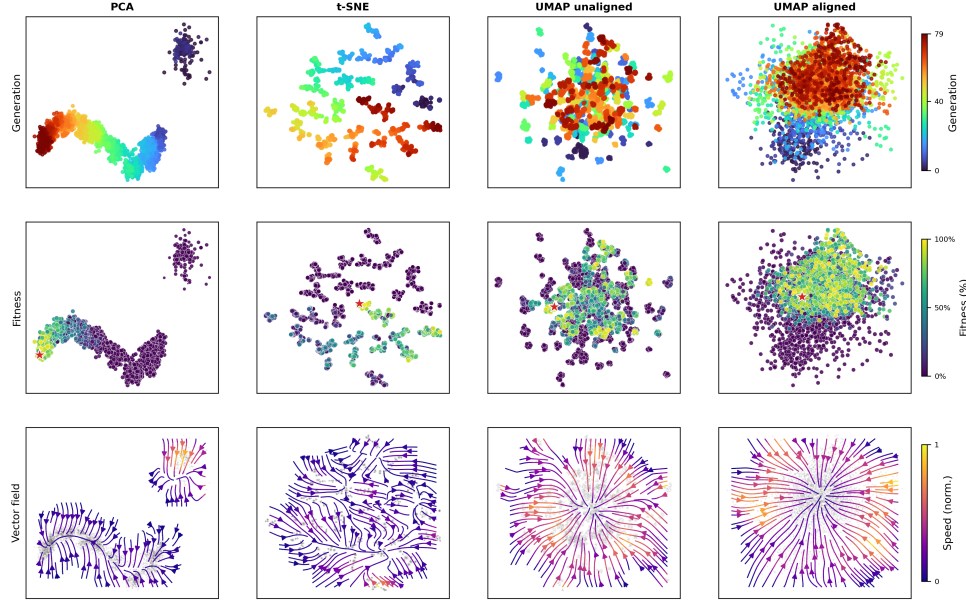


Figure 3: Exp 2: four projection methods on Make Moons (80 generations, 50 individuals). Columns: PCA, t-SNE, UMAP unaligned, UMAP aligned. Rows: generation coloring (top), fitness coloring (middle), velocity field (bottom). Only UMAP aligned ($\lambda = 0.8$) produces interpretable temporal structure and smooth velocity fields.

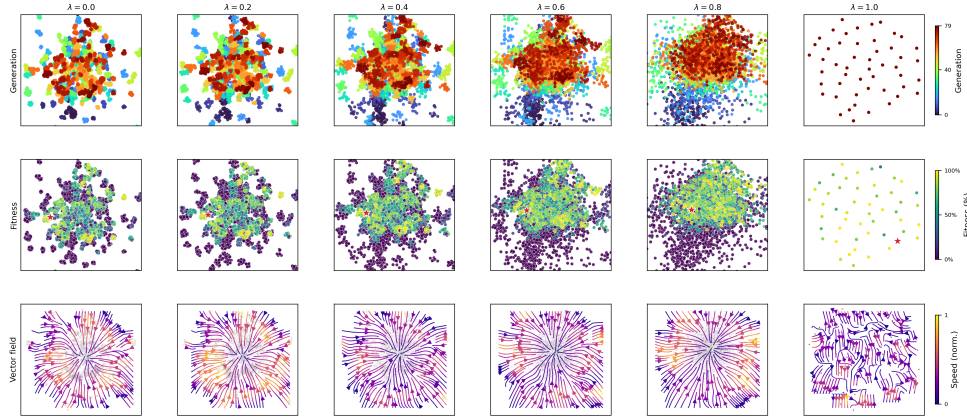


Figure 4: Exp 3: λ sweep ($\lambda \in \{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$, columns). Rows: generation coloring (top), fitness coloring (middle), velocity field (bottom). See Figure 6 for the TC vs. λ curve.

157 4.4 Exp 4: Synthetic Validation

158 To verify that UMAP-detected structure reflects genuine data geometry rather than projection ar-
 159 tifacts, we apply our pipeline to three synthetic datasets with known structure (Table 4): a single
 160 Gaussian cluster, four well-separated Gaussian clusters, and a random walk (each generation inde-
 161 pendently sampled from $\mathcal{N}(0, I)$, no temporal correspondence between points).

162 The random walk yields near-zero temporal coherence ($TC = 0.03$), confirming that independent
 163 point clouds produce no spurious structure. The four-Gaussian dataset achieves high coherence
 164 ($TC = 0.91$), indicating that genuine multi-cluster structure is faithfully preserved across gener-
 165 ations. The single Gaussian shows intermediate coherence ($TC = 0.29$); this reflects UMAP’s
 166 tendency to produce arbitrary rotations between independent projections of structureless data—even
 167 when the underlying point cloud is temporally identical, successive projections rotate relative to each
 168 other, reducing the Pearson correlation in our TC metric. The four-Gaussian case avoids this because

Table 4: Exp 4: synthetic validation. Temporal coherence computed on unaligned embeddings ($\lambda = 0$) to avoid inflation from the alignment procedure itself.

Dataset	Expected modes	Temporal coherence
Single Gaussian	1	0.29
Four Gaussians	4	0.91
Random walk	0	0.03

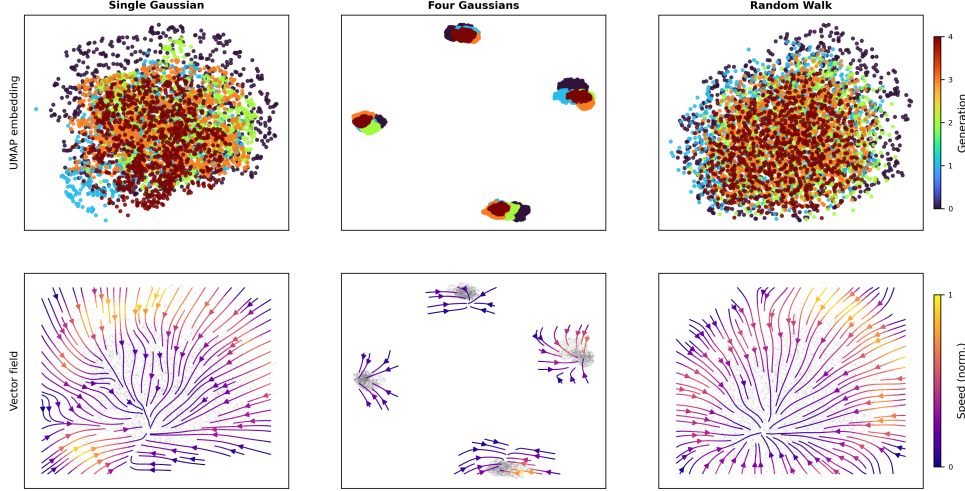


Figure 5: **Exp 4: synthetic validation of UMAP structure recovery.** Columns: Single Gaussian (left), Four Gaussians (center), random walk (right). Rows: UMAP embedding (top), velocity field (bottom). *Single Gaussian*: structureless data yields a diffuse, rotation-prone embedding (TC = 0.29). *Four Gaussians*: four well-separated clusters are faithfully recovered (TC = 0.91). *Random walk*: independently sampled point clouds produce no coherent structure (TC = 0.03), confirming absence of spurious UMAP artifacts.

cluster identity provides anchor points that constrain rotation. Note that Table 4 reports TC at $\lambda = 0$ by design, to isolate UMAP’s behaviour without alignment. With $\lambda = 0.8$ (used in Experiments 1–3), the reference-anchored interpolation suppresses between-generation drift for all datasets including the Single Gaussian, confirming that alignment works as intended across structural regimes. The high TC values in Experiments 1–3 are therefore not inflated by alignment masking structureless data: when there is no structure, $\lambda = 0$ already gives near-zero TC (random walk: 0.03), and alignment cannot manufacture coherence that is not present in the embeddings. Together, these results confirm that the structures observed in the neuroevolution experiments are not UMAP artifacts.

5 Discussion

Our results suggest that neuroevolutionary populations exhibit structured geometric dynamics in weight space that are not captured by scalar fitness metrics. The consistent attractor-like regions and coherent flow patterns across independent seeds indicate that the observed structure is a robust property of the search process, not an artifact of initialization or projection.

The task-dependent multimodality in claim (iii)—compact near-unimodal flow for Make Moons ($d_w = 48$, hidden=16) versus persistent multimodal structure for CIFAR-10 ($d_w \approx 197,000$, hidden=64)—is consistent with known properties of high-dimensional loss landscapes: larger networks have more saddle points and flat regions [4], which could support coexistence of multiple fitness basins simultaneously. However, we caution that the two benchmarks also differ in mutation rate ($\sigma = 0.11$ vs. 0.05), architecture depth, and task complexity; disentangling which factor drives the multimodality difference requires controlled experiments (Section 6).

189 The spatial localization of high-fitness individuals (claim ii) suggests that fitness landscape curvature
 190 is reflected in embedding geometry, which could enable early detection of promising regions before
 191 fitness curves have converged. This has potential implications for adaptive population management
 192 strategies.

193 **Possible mechanisms for the multimodality difference.** Several non-exclusive mechanisms
 194 could contribute to the observed difference in population geometry between benchmarks. First, per-
 195 mutation symmetry: any permutation of hidden neurons produces a functionally equivalent network
 196 at a different weight vector, creating exponentially many symmetric optima that grow with network
 197 size. Second, mutation scaling: smaller perturbations ($\sigma = 0.05$ for CIFAR-10 vs. $\sigma = 0.11$ for
 198 Make Moons) relative to weight magnitude may reduce basin mixing. Disentangling these mecha-
 199 nisms is left for future work.

200 **Practical implications.** The ability to observe convergence basins and multimodal structure in
 201 real time suggests several concrete applications. Practitioners could use velocity fields to detect pre-
 202 mature convergence early, triggering diversity-restoring interventions before fitness plateaus. The
 203 spatial localization of high-fitness individuals could guide offspring allocation in non-uniform mu-
 204 tation strategies, concentrating exploration around identified attractor regions. More ambitiously,
 205 if embedding geometry at generation k predicts eventual convergence targets, it could enable early
 206 stopping or adaptive population sizing.

207 **Motivation.** The case for analyzing neuroevolutionary search at the population level is reinforced
 208 by recent evidence that evolutionary optimization of neural networks is not merely a slower, gradient-
 209 free analogue of stochastic gradient descent but a qualitatively distinct procedure that can produce
 210 qualitatively distinct solutions. Lehman et al. [13] show that evolution strategies optimize for the
 211 average reward of the entire perturbed population rather than the reward gradient at a point, channel-
 212 ing search into regions of parameter space that yield networks with different properties (in their case,
 213 greater robustness to perturbation) than those reached by policy gradients on the same task. More
 214 recently, Kumar et al. [12] argue that conventional SGD tends to learn *fractured*, *entangled* internal
 215 representations, whereas open-ended evolutionary search can yield more unified, factored represen-
 216 tations even when the two procedures match on output behavior. Together, these results suggest
 217 that population-based neuroevolution explores the loss landscape in ways that gradient-based train-
 218 ing does not, and that the relevant structure—multimodality, basin coexistence, robustness-seeking
 219 flow—lives in the geometry of the population, not in any single network’s trajectory. Scalar fitness
 220 curves cannot resolve this geometry. The visualization methodology we develop here is intended
 221 as a step toward a tooling gap that, given how differently neuroevolution behaves from SGD, seems
 222 increasingly worth closing.

223 6 Limitations

224 **UMAP stochasticity.** UMAP embeddings depend on random initialization and neighbor graph
 225 construction. While our temporal alignment reduces between-generation drift, independent runs
 226 with different random seeds may produce qualitatively different embeddings for the same population.
 227 We mitigate this by using a fixed random seed for all projections and reporting results across 5
 228 evolutionary seeds.

229 **Embedding spread as a proxy for multimodality.** We use embedding spread (σ_1) as the primary
 230 quantitative indicator of population structure. While spread correlates with multimodality visually,
 231 it is a coarse proxy: a dispersed embedding could reflect genuine multi-basin structure or simply
 232 a slower-converging population. The velocity field visualizations provide the primary evidence for
 233 claim (iii); spread serves as a reproducible scalar summary.

234 **Scale of benchmarks.** Our experiments use small populations and networks to keep compute
 235 tractable. It is unclear whether the observed phenomena scale to the population sizes and archi-
 236 tectures used in state-of-the-art neuroevolution.

237 **Mutation scaling confound.** The two benchmarks use different mutation rates ($\sigma = 0.11$ for Make
 238 Moons, $\sigma = 0.05$ for CIFAR-10), chosen empirically to produce comparable per-generation fitness
 239 progress on each task. This confounds the dimensionality contrast in claim (iii): a smaller σ relative
 240 to weight magnitude may itself reduce cross-basin mixing, independently of dimensionality. A

241 controlled experiment varying σ within a single benchmark while holding architecture fixed would
242 help disentangle these effects.

243 **Methodological scope.** The vector-field-on-aligned-projections methodology is established in the
244 context of single-network hidden activation visualization [1]; our contribution is in applying it
245 to evolving weight *populations* and characterizing the distinct population-level phenomena that
246 emerge, not in the projection or vector field machinery itself.

247 **Causal interpretation.** We observe *correlations* between embedding structure and fitness dynamics.
248 We do not claim that the geometric structure *causes* or *predicts* evolutionary outcomes; only that it
249 is consistently present and visually informative.

250 7 Related Work

251 **Neuroevolution.** Modern neuroevolution methods [26, 24, 23] have achieved strong performance
252 on RL benchmarks but remain poorly understood mechanistically. Our work is complementary
253 to fitness landscape analysis [10] and quality-diversity methods [18] that also seek to understand
254 population structure.

255 **Vector field visualization of projection trajectories.** Cantareira et al. [1] introduced the method-
256 ology of constructing vector fields from sequentially aligned projections to visualize hidden layer
257 activity in deep networks across training epochs and layers. Their approach combines a generic
258 projection alignment model [2] with Vector Field K -Means [6] for trajectory aggregation; Rauber
259 et al. [22, 21] established related trajectory-based visualization of hidden activations using dynamic
260 dimensionality reduction. Our work applies this family of techniques to a qualitatively different
261 regime: instead of tracking a single network’s internal states, we track *populations* of independent
262 networks across evolutionary generations. This shift introduces phenomena absent in single-network
263 visualization: selection pressure creates fitness-correlated spatial structure, attractor competition
264 emerges from parallel independent trajectories, and task-dependent search dynamics manifest at the
265 population level rather than within individual activation patterns.

266 **Weight space visualization.** Li et al. [15] visualize loss landscapes via filter-normalized linear inter-
267 polation; Goodfellow et al. [7] study loss surfaces along linear paths. These approaches characterize
268 a single network’s loss surface; our work characterizes the collective trajectory of a population. For
269 a survey of visual analytics in deep learning, see Hohman et al. [9].

270 **Dimensionality reduction for neural networks.** UMAP and t-SNE have been applied to visual-
271 ize neuron activations [3] and model families [20]. Closer to our temporal setting, Grisci et al. [8]
272 propose *weighted t-SNE*, a feature-importance-weighted variant of t-SNE that can be applied at suc-
273 cessive training epochs to visualize how a single neural network reshapes its data representation as
274 the learned feature scores evolve. Our work differs in two key respects: (i) we track a *population*
275 of independent networks rather than a single model, so the structures we observe (multimodality, at-
276 tractor competition, fitness localization) emerge from selection pressure across the population rather
277 than from gradient-based learning of one network’s feature weights; and (ii) our temporal axis is
278 evolutionary generations rather than training epochs, with no per-individual continuity of paramet-
279 ers across selection events. Our contribution is therefore to apply temporally aligned projection to
280 evolving weight *populations*, revealing population-level geometric phenomena specific to the evolu-
281 tionary regime.

282 **Behavioral diversity in evolution.** MAP-Elites [18] and novelty search [14] explicitly maintain
283 behavioral diversity; our approach provides post-hoc geometric analysis of diversity in weight space
284 without requiring behavioral descriptors.

285 8 Conclusion

286 We have presented an analysis methodology for neuroevolutionary population dynamics based on
287 temporally aligned UMAP embeddings and population velocity fields. Applied across benchmarks
288 with qualitatively different search regimes, the method consistently reveals geometric structure—
289 coherent flow, fitness localization, and task-dependent multimodality—that is invisible in scalar
290 fitness metrics. Ablation and synthetic validation experiments confirm that the observed structure
291 reflects genuine evolutionary dynamics rather than projection artifacts. We release code and interac-

292 tive visualizations to facilitate adoption by the neuroevolution community (<https://anonymous.4open.science/r/neurips-2026-9729>).
293

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A Additional Experimental Details

A.1 Network Architecture and Training

All networks are one-hidden-layer MLPs with sigmoid activations. Make Moons: input $2 \rightarrow 16 \rightarrow 1$ (sigmoid output, binary cross-entropy). CIFAR-10: input $3072 \rightarrow 64 \rightarrow 10$ (softmax output, negative log-likelihood). Elite fraction = 20%, minimum elite = 2 individuals.

A.2 Compute Resources

All experiments run on a single CPU (Apple M-series, 16GB RAM). Approximate wall-clock times: Make Moons (5 seeds) $\approx$ 1 min; CIFAR-10 (5 seeds) $\approx$ 20 min; Exp 2–3 $\approx$ 1–2 min each; Exp 4 $\approx$ 1 min. No GPU required. Total reproduction time $\lesssim$ 30 min.

A.3 Full Multi-Seed Results

Table 5 reports all per-seed metrics including attractor count, estimated by DBSCAN with adaptive ε (90th percentile of k -NN distances, $k = 5$). Attractor counts are sensitive to embedding density and should be interpreted as rough ordinal indicators rather than precise geometric measurements; we use embedding spread as the primary metric in the main text.

Table 5: Per-seed metrics for all benchmarks. Attractor count uses DBSCAN with adaptive ε ; values vary with embedding density and are provided for completeness only. $\dagger$ CIFAR-10 seed 31 reports convergence at generation 3; the fitness trajectory began close to its final value under this initialization, so 95% of the negligible improvement span ($\bar{f}_0 \approx \bar{f}_T$) was reached trivially early. This is a numerical artifact of the convergence-rate definition when the fitness is nearly flat, not an unusually fast run; it is the primary driver of the high convergence variance (± 10) reported in Table 2.

Benchmark	Seed	Spread σ_1	Fitness	Conv. (gen)	Attractors
Make Moons	42	1.88	-0.169	57	19
	123	1.92	-0.171	41	13
	7	1.92	-0.163	47	13
	31	2.07	-0.175	45	13
	99	2.03	-0.179	45	18
CIFAR-10	42	2.45	-2.464	19	5
	123	2.31	-2.497	26	4
	7	2.79	-2.497	26	5
	31 †	2.64	-2.519	3	5
	99	2.89	-2.472	24	4

376 A.4 Hyperparameter Table

Table 6: Full hyperparameter table for all benchmarks.

Benchmark	Pop	Gens	Hidden	σ	λ	PCA dims
Make Moons	50	80	16	0.11	0.8	—
CIFAR-10	20	40	64	0.05	0.8	50

377 A.5 Temporal Coherence vs. Alignment Strength

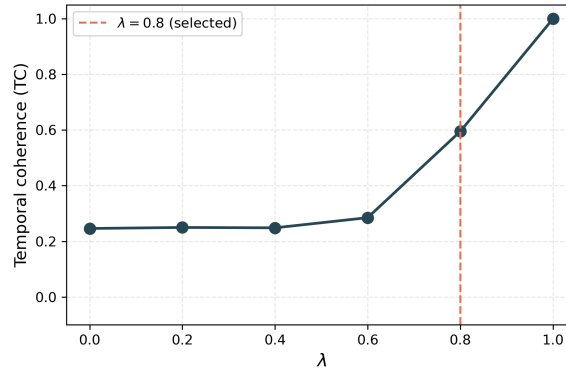


Figure 6: Temporal coherence as a function of alignment strength λ . Coherence increases monotonically; $\lambda = 0.8$ (dashed line) provides the best trade-off between coherence ($TC = 0.587$) and embedding diversity ($\sigma_1 = 1.88$). At $\lambda = 1.0$, all generations collapse to the reference ($TC = 1.0$) but temporal structure is destroyed.

378 A.6 Interactive Visualization Tool

379 We developed a browser-based interactive tool built with Streamlit that exposes all visualization
380 components described in this paper—aligned UMAP projections and population velocity fields—
381 through a parameter-driven interface. Users can configure population size, number of generations,
382 hidden dimensionality, mutation rate, alignment strength λ , and toggle between static (Matplotlib)
383 and interactive (Plotly) rendering modes in real time.

384 Figures 7–9 show both rendering modes on the Make Moons benchmark with $\lambda = 0.8$ and 80
 385 generations.

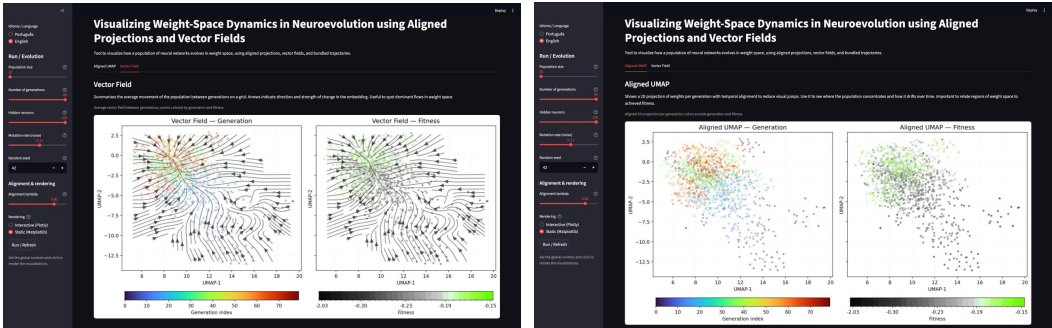


Figure 7: Static Matplotlib mode. **Left:** Vector Field tab—population flow colored by generation (left panel) and fitness (right panel). **Right:** Aligned UMAP tab—per-generation 2D embeddings with generation and fitness colorbars. The sidebar exposes all run parameters for rapid hypothesis testing.

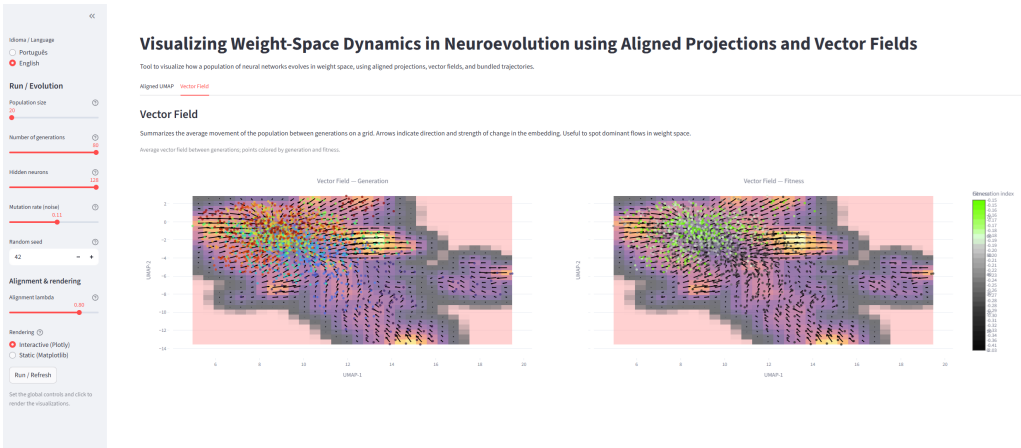


Figure 8: Interactive tool (Plotly mode) showing the Vector Field tab on Make Moons, $\lambda = 0.8$, 80 generations. Quiver arrows indicate population flow direction and magnitude; background density heatmap shows where individuals concentrate. Left: colored by generation index. Right: colored by fitness. The sidebar exposes all run parameters for rapid hypothesis testing.

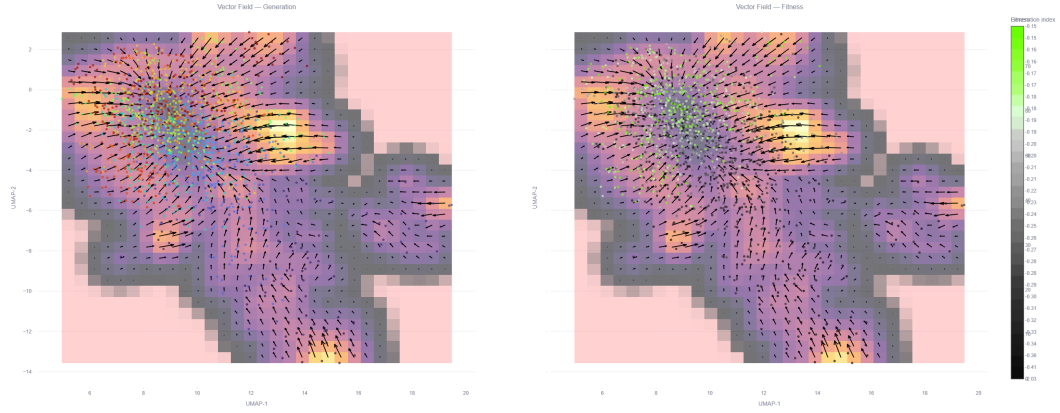


Figure 9: Detail view of the interactive Plotly quiver vector field. The density background and directional arrows together reveal the dominant convergence basin and the multimodal structure of the Make Moons fitness landscape. Points are individually hoverable; the plot supports zoom and pan.

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